
Introduction to Economics

Class 1 — Why economics, why now

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Universitat de Barcelona · Group A6

14 September 2026

Who I am

- Studied in Pilsen, Kaohsiung, Barcelona and New York
- Mathematics first, then economics — PhD, New York University, 2019
- Across the Atlantic by sailboat, twice
- Now: the economics of AI safety and governance

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Plan for today

10:00	The course
10:20	You — introductions in groups
10:50	Warm-up quiz
11:00	Break
11:10	The answers, and a pair task
11:40	What is economics?

The course in one slide

1	Introduction to economics	Ch. 1, 2, 17, 20
2	Supply and demand	Ch. 3
3	Efficiency and intervention	Ch. 6–9, 17
4	Firms and market structure	Ch. 5, 10–13
5	Measuring a country's income	Ch. 20
6	Cost of living and unemployment	Ch. 20, 22
7	The open economy	Ch. 21, 25, 33

Mankiw & Taylor, *Economics*, 5th ed. · CORE, *The Economy* — free online

How you are graded

Midterm 1	30 %
Midterm 2	30 %
In-class quizzes	10 %
Problem sets	10 %
Project — poster	20 %

Pass: midterm average ≥ 5 and at least 4 in each $\rightarrow$ no final exam.

Project: groups of 2–3, one poster, the models applied to the world around you. Today's quiz does not count.

Practicalities

- Campus Virtual — materials from Thursday
- Email: pakoinf@gmail.com
- Weekly rhythm and practical sessions: details on Thursday
- Teaching language: English

Using AI

- Quizzes and midterms: paper only, no AI
- Problem sets: use it, say where
- Project: use it, log it, check it
- Hidden AI in graded work: not allowed

The next two weeks

Thu 17 Sep	Next class — Chapter 1
Mon 21 Sep	No class
Thu 24 Sep	La Mercè — holiday
Mon 28 Sep	Block 1 in detail
Midterms	Dates announced in class

Introduce yourselves — in groups of four

1. Your name
2. Where you are from
3. Why economics
4. One thing you hope to get from this course
5. One question you have about it

About five minutes each.

Then, as a group

- Agree on your **top two hopes**
- One card: the two hopes, your four names
- A few groups read theirs out

Warm-up quiz

- Eight questions — **not graded**
- One capital letter per box
- Your name on the sheet
- Hand it in before the break

Guessing is allowed and expected.

Break

Back at 11:10

The answers

Eight questions, eight stories

Question 1

For thousands of years, income per person barely changed. Then it started rising steadily and has not stopped. Where and when did this take-off first happen?

- A)** Ancient Rome, around 100 AD
- B)** Italy, around 1400 (the Renaissance)
- C)** Britain, around 1750–1800
- D)** The United States, around 1950

Question 1

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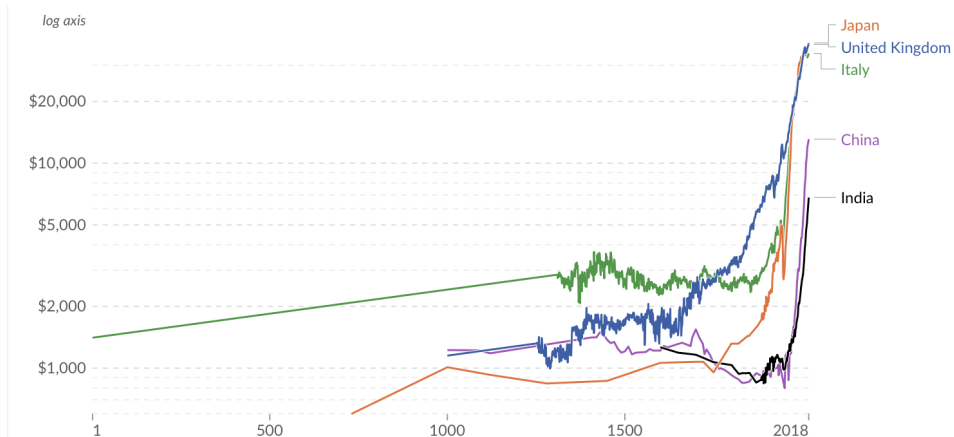
C) Britain, around 1750–1800



D) The United States, around 1950

CORE, *The Economy*, Unit 1.

Flat for centuries, then straight up



Britain first, from around 1760 — then the others, one by one.

Real GDP per person, log scale, 1–2018. CORE, *The Economy*; Maddison Project Database.

Question 2

Britain's population quadrupled between 1800 and 2000. Over the same 200 years, the real wage of a British worker...

A) fell by half, as Malthus predicted

B) barely moved

C) roughly doubled

D) rose about ninefold

Question 2

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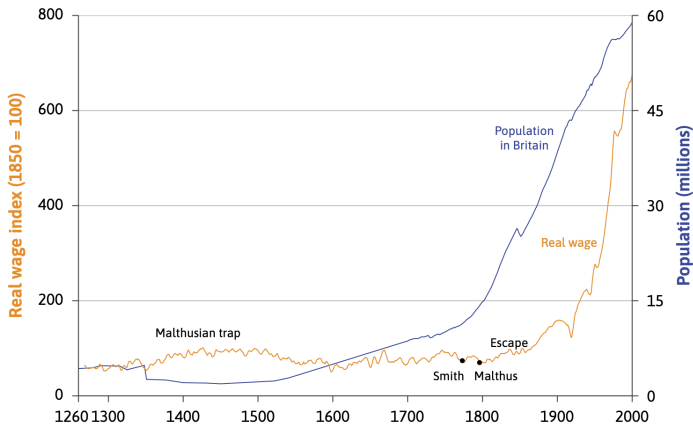
C) roughly doubled

D) rose about ninefold



CORE, *The Economy*, Unit 1.

Wages rose ninefold while population quadrupled



Malthus published in 1798 — just as the escape began.

Britain, 1260–2000. CORE, *The Economy*.

Question 3

Since about 2010, the computing power used to train the largest AI models has grown by roughly how much per year?

A) 4 times

B) 1.4 times

C) 10 per cent

D) 100 times

Question 3

Since about 2010, the computing power used to train the largest AI models has grown by roughly how much per year?

A) 4 times



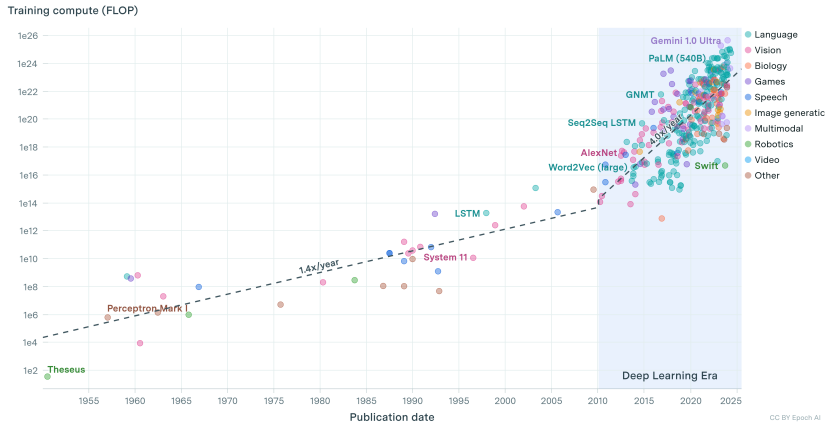
B) 1.4 times

C) 10 per cent

D) 100 times

Epoch AI, *Notable AI models*.

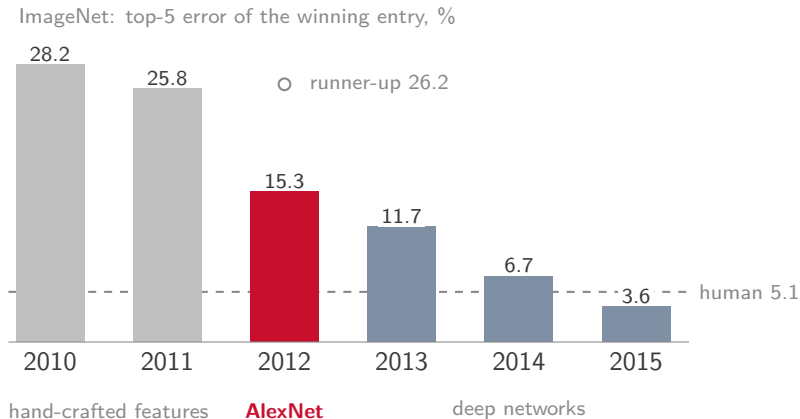
Four times a year, for fifteen years



$1.4\times$ a year before 2010, about $4\times$ since — a millionfold in a decade.

Training compute of notable AI models, log scale. Epoch AI.

2012: one deep network beats the field by ten points



Two GPUs, 1.2 million labelled images, one deep network.

ILSVRC results; Krizhevsky, Sutskever & Hinton (2012); Russakovsky et al. (2015).

Question 4

In 2023 the best AI models could complete, with 80 % success, software tasks that take a person about 40 seconds. By April 2026, the frontier models could handle tasks of about. . .

A) 3 hours

B) 20 minutes

C) 2 days

D) 2 minutes

Question 4

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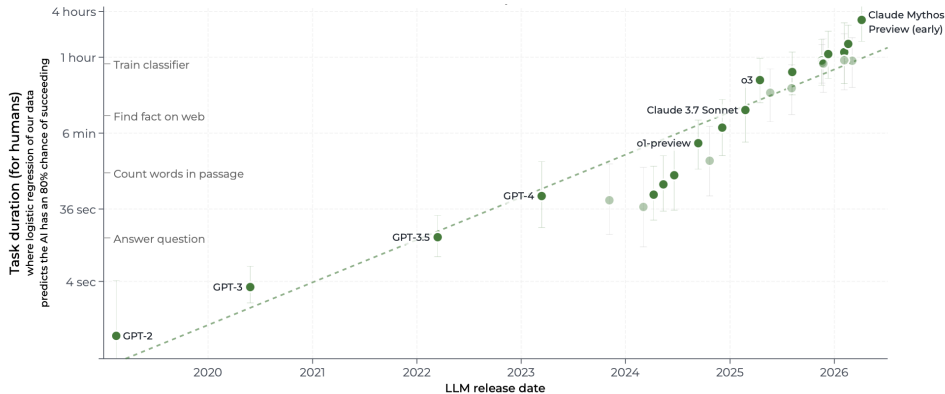
B) 20 minutes

C) 2 days

D) 2 minutes

METR, *Time horizon of software tasks.*

From 40 seconds to 3 hours in three years



40 seconds in March 2023, about 3 hours by April 2026 — doubling every few months.

Software tasks completed 80 % of the time, by model release date. METR.

Question 5

British textile workers smashed the new machines they blamed for their lost wages. When?

A) 1650

B) 1811

C) 1871

D) 1926

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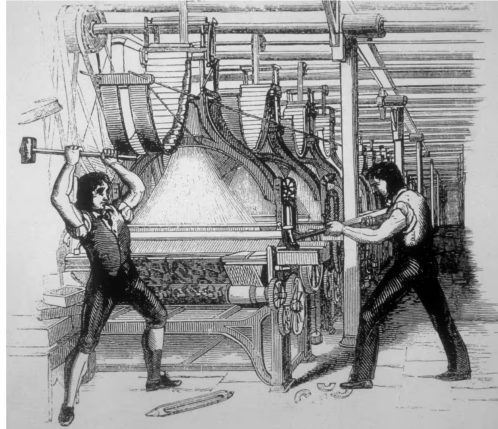


C) 1871

D) 1926

The Luddite risings, 1811–1812.

1811 — half a century after the first machines



Frame-breaking in Nottinghamshire and Yorkshire, 1811–1812.

API / Gamma-Rapho / Getty Images.

Question 6

When power looms arrived in British factories, the number of operatives the factories needed. . .

- A)** fell to almost zero
- B)** fell by half
- C)** rose, because each loom needed an operative
- D)** stayed the same

Question 6

When power looms arrived in British factories, the number of operatives the factories needed. . .

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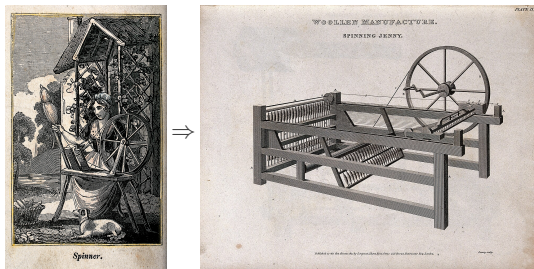
C) rose, because each loom needed an operative ✓

D) stayed the same

British cotton factories, nineteenth century.

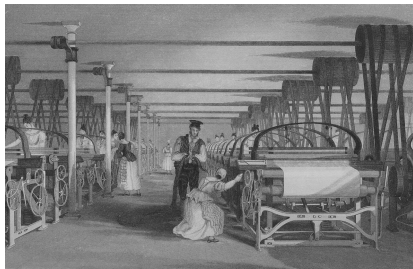
Same industry, opposite answers

Spinning: machines *for* workers



One jenny did the work of many spinners.

Weaving: machines *need* workers



Every loom needed an operative.

Engravings: a spinning jenny, 1811; a power-loom weaving shed, 1830s.

Question 7

What is the true cost to you of sitting in this class right now? (Think like an economist.)

- A)** Nothing — it is already paid for
- B)** The best alternative use of these two hours
- C)** Only the metro ticket you paid to get here
- D)** The tuition fee you already paid

Question 7

What is the true cost to you of sitting in this class right now? (Think like an economist.)

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Opportunity cost — Chapter 1, on Thursday.

Question 8

Economists have won Nobel Prizes for studying. . .

- A)** why some countries are rich and others poor
- B)** how to design auctions for mobile-phone licences
- C)** marriage, crime and discrimination
- D)** all of the above

Question 8

Economists have won Nobel Prizes for studying. . .

- A) why some countries are rich and others poor
- B) how to design auctions for mobile-phone licences
- C) marriage, crime and discrimination
- D) all of the above** ✓

2024 Acemoglu, Johnson, Robinson · 2020 Milgrom, Wilson · 1992 Becker.

Ten years back, ten years forward

- The biggest advances of the past ten years?
- What do you expect from the next ten?
- In pairs. Two lines on a card. Ten minutes.

The seed of your poster project.

What is economics?

From the answers to the ideas

Three questions every society answers

What gets produced?

How is it produced?

Who gets it?

Your turn — pick one case

What, how, and for whom — for one of these:

- Who gets a place at UB
- Who gets a ticket to a sold-out concert
- Who does the dishes at home

In pairs. Six minutes. Then we hear a few.

The words for what you just did

scarcity

choice

opportunity cost

incentives

trade

Chapter 1, in detail, on Thursday.

Before Thursday

- Read 1.1 “What is economics” — Mankiw & Taylor, Ch. 1
- Next class: Thursday 17 September, 10:00
- Optional: ask a grandparent what a day looked like at your age.
One sentence, for 28 September

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